

RDO ONE

Dual-Exchange Perpetuals Terminal

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What Is RDO ONE?

RDO ONE is an advanced trading terminal for decentralised perpetual futures. It connects directly to two cutting-edge DEX protocols — **Hyperliquid** and **Aster DEX** — inside a single, unified interface. Think of it as a Bloomberg terminal for on-chain derivatives: live charts, real-time order books, instant order placement, portfolio tracking, cross-chain fund transfers, and a crypto news feed — all in one black-screen application.

The core idea

Most traders have to open 3–4 different websites to manage their positions across different DEXes. RDO ONE collapses everything into one terminal that looks and feels like a professional trading platform, not a crypto DApp.

Why perpetuals?

Perpetual futures ("perps") are the most-traded instrument in crypto. Unlike spot trading you don't need to own the underlying asset — you trade price direction with leverage. The daily volume across all DEX perp protocols exceeds \$10 billion. RDO ONE targets power users who need professional tools to work with that market.

The two modes

BASIC — Hyperliquid

Hyperliquid is the #1 on-chain perp protocol by volume. It runs its own high-performance L1 blockchain and offers 100+ perpetual markets with extremely low fees and no gas costs for trading. BASIC mode gives you a professional terminal to trade on Hyperliquid.

EXTRA — Aster DEX

Aster DEX is a newer perpetuals exchange with deep liquidity and up to 200× leverage. It runs on its own chain and focuses on capital efficiency. EXTRA mode brings Aster into the same terminal so you can switch instantly without opening a new browser tab.

The Five Pages

RDO ONE is split into five dedicated areas. Each serves a specific purpose in the trading workflow.

1. Trade

The main terminal — where you actually place orders

- Live candle chart powered by TradingView
- Real-time order book with bid/ask depth
- Live trade feed — every executed trade
- Order panel: Market and Limit orders
- Buy/Long and Sell/Short with one click
- TP / Stop Loss configuration
- Position sizing slider (25% / 50% / 75% / 100%)
- Bottom panel: open positions, balances, order history

2. Markets

Global crypto market overview

- Scrolling ticker with prices for top 50 pairs
- Total market cap and 24h trading volume
- BTC dominance percentage
- Fear & Greed index (market sentiment)
- Top trending coins right now
- Top 24h gainers and losers
- Live price converter between any two assets

3. News

Crypto news aggregated from 8 sources

- Sources: CoinDesk, Decrypt, The Block, BeInCrypto, CryptoSlate, and more
- Category filters: DeFi, Bitcoin, Ethereum, Altcoins, Regulation, NFT
- Source filter chips — show/hide individual publications
- Auto-refreshes every 5 minutes
- Sorted by publication time, newest first

4. Portfolio

Track your wallet's performance

- Works for both Hyperliquid and Aster DEX wallets
- Enter any wallet address — no login needed
- Total realised PnL, win rate, number of trades
- Cumulative PnL chart over time
- Daily PnL calendar heatmap
- Full list of closed and open positions
- Live account equity, margin used, leverage

5. Transfer

Move funds across chains with zero friction

- Withdraw — move funds from HL/Aster to any EVM chain
- Send — cross-chain token transfers (wallet A → wallet B)
- Between Accounts — move between HL and Aster accounts
- Powered by LI.FI — finds the best bridge route automatically
- No need to find a bridge manually

How It Works

You don't need to be a developer to understand the architecture. Here's what happens under the hood when you use RDO ONE.

The browser does everything sensitive

RDO ONE is built on a fundamental security principle: **your private key and wallet never leave your browser**. When you place an order, the browser asks your MetaMask or Rabby wallet to sign the trade. The signature — not your key — is then sent to the exchange. This is the same mechanism used by Uniswap and every major DeFi protocol.

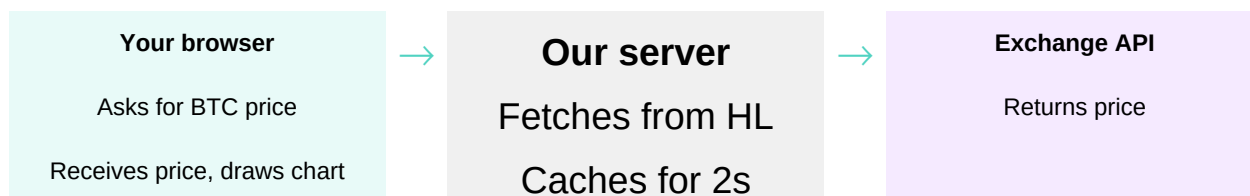
Non-custodial

The RDO ONE team can never access your funds. There is no account, no email, no password. You connect with your existing crypto wallet and trade directly on-chain.

Why is there a backend server?

Web browsers have a security restriction called CORS that prevents websites from fetching data from external APIs directly. If you visit rdoone.com, your browser won't let the page fetch prices directly from Hyperliquid's API — it gets blocked.

Our backend server sits in the middle as a **relay**. It fetches the data from the exchanges on the browser's behalf, caches it for a few seconds so we don't hit rate limits, and delivers it to the app. The server never sees your wallet or private key.



Real-time data via WebSocket

Prices update in real-time — not every few seconds, but every time a trade happens on the exchange. This is done through a technology called WebSocket, which keeps a permanent open connection between the app and the exchange. When BTC trades at a new price, that update arrives in milliseconds and the candle chart updates instantly.

Cross-chain transfers via LI.FI

Moving tokens between different blockchains normally requires finding the right bridge, waiting for confirmations, and paying multiple gas fees. LI.FI is an aggregator that automatically finds the cheapest and fastest route across all major bridges (Stargate, Across, Hop, etc.) and executes it in a single transaction from your wallet. The Transfer page in RDO ONE is powered entirely by LI.FI.

Tech Stack at a Glance

Here's a plain-English breakdown of the technologies we're using and why.

Technology	What it is	Why we use it
Vite	Frontend build tool	Blazing fast dev server, instant hot reload
Vanilla JS	Plain JavaScript, no framework	Zero overhead, full control, tiny bundle size
Lightweight Charts	TradingView's open-source chart library	Professional candle charts with real-time updates
ethers.js	Ethereum wallet library	Connects to MetaMask, signs orders with EIP-712
WalletConnect v2	Mobile wallet protocol	Lets users connect any mobile wallet (Trust, Rainbow, etc.)
LI.FI Widget	Cross-chain bridge aggregator	Powers the Transfer page — finds best bridge route
Fastify	Node.js web server	Very fast, minimal overhead, perfect for a proxy/relay
Redis	In-memory cache	Caches exchange data for 2–30s, prevents rate limit hits
WebSocket relay	Persistent connection proxy	Streams live prices from exchanges to all connected users

Design System

RDO ONE uses a Hyperliquid-inspired dark theme with a pure black (#000000) background and a teal accent colour. The buy side uses mint green, the sell side uses pink — both borrowed from the Bloom Manager terminal for a clean, professional feel. All navigation uses a pill-button active style (small dark filled background) consistent across every page in the app.

Current Status

Component	Status	Notes
Trade terminal UI	✓ Done	Charts, order book, live trades, order panel
BASIC mode (Hyperliquid)	✓ Done	REST + WebSocket, order signing, positions
EXTRA mode (Aster DEX)	✓ Done	REST + WebSocket integrated
Markets page	✓ Done	CoinGecko data, ticker, converter
News page	✓ Done	8 RSS feeds, category filters
Portfolio page	✓ Done	PnL charts, positions, trade history
Transfer page	✓ Done	LI.FI bridge for 3 transfer types
Backend proxy server	✓ Done	Fastify + Redis, all proxy routes
Internationalisation	✓ Done	EN / RU / ZH translations
Production deployment	■ Pending	Backend needs to be deployed to Railway / Fly.io
Mobile responsive layout	■ Pending	Terminal is desktop-first; mobile TBD
User accounts / watchlist	■ Planned	Optional — wallet-based persistent settings

How You Can Help

If you're reading this, you might be interested in contributing. Here are the areas where extra hands would make the biggest difference:

Area	What's needed	Skills required
Backend deployment	Deploy and configure the Fastify server on Railway or Fly.io	Node.js, Docker basics, cloud platforms
Aster order signing	Implement HMAC-signed order placement for EXTRA mode	JavaScript, HMAC-SHA256, REST APIs
Mobile layout	Design and build a responsive layout for phones and tablets	CSS, responsive design, UX
Testing	Test order flows on both exchanges end-to-end	Manual QA, ideally some Playwright experience
Design / UI polish	Visual improvements, animations, icon set	CSS, Figma, eye for detail

Get started

The repository is at github.com/wholelottawork/rdo-one-setup. Run `npm install` then `npm run dev` and the terminal is live at `localhost:5176`. No backend needed for most frontend work — the dev server proxies to a shared staging backend.